

Methods

A method is basically a block of code that run when it is called and the parameters can also be passed in it.

They are basically used to reuse code: define the code once, and use it many times.

Methods can be **Pass by reference and Pass by value**

When the value is passed inside the parathesis following the method name then it is called parameter ,while when its value is provided when the function or the method when it is called it is argument.

Note that when you are working with multiple parameters, the method call must have the same number of arguments as there are parameters, and the arguments must be passed in the same order.

In Java, variables are only accessible inside the region where they are created. This is called **scope**.

Pass by value

In Pass by value the copies of the argument is made and so the original value of the variable is not changed

****Pass by reference**

While in Pass by reference ,the value of original variable is changed as it uses pointers to point at the reference or the memory location of the argument

Hence Java is strictly Pass by value ,as Java does not have pointers because it is a vulnerability of accessing the addresses directly and leads to major security issues and these are some of the reasons why it is considered as more ****secure****.

Changes are reflected

When you modify the object's data:

```
obj.name = "Rahul";obj.age = 21;
```

because both references point to the same object.

Changes are NOT reflected

When you change the reference itself:

```
obj = new Student();
```

because only the local copy of the reference changes

Java Recursion

Recursion is the technique of making a function call itself. This technique provides a way to break complicated problems down into simpler problems which are easier to solve.